Dani Roytburg

CURRICULUM VITAE

droytbur@andrew.cmu.edu · djroytburg.github.io · GitHub · Google Scholar

Master's student in the Machine Learning Department at Carnegie Mellon University focused on reliable and interpretable AI, multi-agent training paradigms, and activation-based model analysis.

Education

Carnegie Mellon University, School of Computer Science expected May 2027

M.S. in Machine Learning

Emory University, College of Arts and Science 2025

B.S. in Computer Science; Quantitative and Social Sciences · *summa cum laude*

Research & Professional Experience

ML Alignment and Theory Scholars May 2026 – September 2026

Research Fellow · Berkeley, CA

Advised by Shi Feng. Developed persona-training methods which emulate midtraining for better model organisms and constitutions. Engineered evaluations across 100+ training runs on synthetic-document pipelines. Evaluated how tastefully frontier models can test ideas in safety research.

Carnegie Mellon University September 2025 – Present

Research Assistant, DI Group · Pittsburgh, PA

Develop multi-agent reinforcement learning environments to improve legibility of reasoning traces. Design user studies and adversarial setups to measure influence of legibility on human and multi-agent interactions.

Martian Research May 2025 – February 2026

Research Fellow · San Francisco, CA

Engineered activation profiles of language model evaluators to suppress self-preferential bias, applied to LLM routers (ICML 2026, MechInterp @ NeurIPS 2025). Winner of Mechanistic Router Interpretability Hackathon (40 submissions).

Emory University November 2021 – May 2025

Research Assistant, Digital Humanities Lab · Atlanta, GA

Designed and trained large-scale graph-text models; evaluated adversarial robustness of LLMs; built historical correspondence networks and dashboards.

ZS Associates June 2024 – August 2024

Internship - Business Technology Solutions · Evanston, IL

Automated ETL pipelines and implemented synthetic-data imputation for compromised client data.

Cloverpop January 2024 – May 2024

Intern — Machine Learning · Chicago, IL

Deployed decision-management systems and structured-prediction models on meeting transcripts.

McCormick and Company June 2023 – August 2023

Intern — Global Marketing and Sales Data Science · Hunt Valley, MD

Built Amazon sales data visualizations and automated analytics with GCP BigQuery and BERT-based embedding search.

Northeastern University June 2022 – October 2022

Research Assistant, Mediacloud Lab · Boston, MA

Extended NER to long-tail ethnics cuisines using Bon Appetit data.

Publications

* Equal contribution / co-first authorship.

Under review

Pre-training interventions, ex post facto: Grafting model beliefs across checkpoints
Peter Nutter*, **Dani Roytburg***, Clément Dumas, Jinghua Ou, Shi Feng.
Under review, 2026.
Paper · Blog post · Project website · Code

Conference & workshop papers

Disentangling Models from Personas in Heterogeneous LLM Simulations
Dani Roytburg, Daphne Ippolito.
Second Workshop on Social Simulation with LLMs: Fidelity in Applications (*COLM 2026, Spotlight, Best Paper Award*), 2026.
Paper · Project website · Code

Measuring Weak-to-Strong Legibility of Reasoning Models
Dani Roytburg, Shreya Sridhar, Daphne Ippolito.
Advances in Neural Information Processing Systems (*NeurIPS*), 2026.
Also at: ICML 2026 Workshop on AI for Good (AI4GOOD).
Paper

Are LLM Evaluators Really Narcissists? Sanity Checking Self-Preference Evaluations
Dani Roytburg, Matthew Bozoukov, Matthew Nguyen, Jou Barzdukas, Mackenzie Puig-Hall, Narmeen Oozeer.
Forty-Third International Conference on Machine Learning, 2026.
Paper · Code

Mind the Gap: Pathways Towards Unifying AI Safety and Ethics Research
Dani Roytburg*, Beck Miller*.
Proceedings of the International Association for Safe and Ethical AI, 2026.
Paper · Code

Breaking the Mirror: Activation-Based Mitigation of Self-Preference in LLM Evaluators
Dani Roytburg*, Matthew Bozoukov*, Hongyu Fu, Matthew Nguyen*, Jou Barzdukas*, Narmeen Fatimah Oozeer.
Mechanistic Interpretability Workshop at NeurIPS 2025, 2025.
Also at: NeurIPS 2025 Workshop on Evaluating the Evolving LLM Lifecycle; NeurIPS 2025 Workshop on Reliable ML from Unreliable Data.
Paper · Code

Words and Action: Modeling Linguistic Leadership in # BlackLivesMatter Communities
Dani Roytburg*, Deborah Olorunisola*, Sandeep Soni, Lauren Klein.
Proceedings of the International AAAI Conference on Web and Social Media, 2025.
Paper

Awards

Best Paper Award, COLM Social Simulation Workshop	2026
Best Poster, LTI Student Research Symposium	2026
Emory University Dean's List	2022, 2024, 2025
Winner, Martian Research Mechanistic Interpretability Hackathon	2025

Technical Skills

Python · Java · R · JavaScript · Typescript · SQL · PyTorch · JAX · HuggingFace Transformers · scikit-learn · spaCy · networkx · D3.js · React.js · GCP · Docker · MySQL